Xinyu Sun

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EDUCATION

ETH Zurich 2020.09 – 2024.02
Robotics System and Control, D-MAVT Zurich, Switzerland
Overall GPA: 5.62/6

Shanghai Jiao Tong University 2016.09 - 2020.06
School of Mechanical Engineering, Tsien Hsue-Shen Honor Program Shanghai, China
Overall GPA: 90/100

SKILLS

Language ability: English, Chinese, German
Computer skills: C#, C++, Python, Git, Latex

Work

Driving Global Scalability of Industrial Applications 2024.06 – now Agile Robots Munich

- Pioneering an industrial "copilot", allowing scalable customer support without senior engineer travel.
- Scaled AgileX, a global robotics application platform, by enabling hardware integration, workflow orchestration, and UI-driven interaction; successfully deployed across China, Japan, and India.
- Achieved 0.5mm operational accuracy in real-world deployments through the RL-based methods.
- Built foundational vision components to enable robust and precise manipulation tasks.

Aerial Vehicles Simulation for Autonomous Industrial Inspection 2023.09-2024.02 Flyability Lausanne

- Implemented drone dynamics within ISAAC-SIM.
- Simulated the property of lidar, camera and IMU, creating 10 industrial scenes.
- Integrated the simulator into 3 repositories, conducting tests on their mapping and navigation code.

Human Body Parameter Registration from Silhouette 2022.05-2022.09 Philips Eindhoven

- Preprocessed 50k raw meshes and augmented them with pose changes.
- Built and trained a model fusing silhouette and meta data to PCA coefficients.
- Tested on photos from reality and predicted breast/waist girths within 25mm error.

PROJECT

Safe Reinforcement Learning for Multi-player Drone Racing 2022.11-2023.06 UZH RPG

- Generated competitive and diverse opponent strategies by adversarial attack and league play.
- Improved sampling efficiency by incorporating suboptimal racing strategies.
- Addressed data imbalance through task divisions and training separation.
- Increased survival rate from 45% to 70% compared to baselines.

Conditional Predictive Reinforcement Learning for Driving Policy 2021.09-2022.05 ETHZ CVL

- Established a RL framework with reward shaping in the CARLA simulator.
- Achieved a 90% accuracy in predicting traffic-related events using visual features.
- Tested in the NoCrash benchmark, achieving an impressive 85% route completion rate.

Deep Learned Constellation Descriptors for Object-based Map 2021.03-2022.03 ETHZ ASL

- Developed a DeepGCN descriptor of simulation constellations by Max-Relative Graph Convolution.
- Created a training dataset generator to simulate object distribution in urban scenarios.
- Constructed a localization benchmark, incorporating RANSAC, to evaluate descriptors.
- Achieved above 80% successful localization within 2m error while 60% by baselines on real datasets.